

UTSIRA FYR, Norway

WMO: 014030

Lat: 59.306N

Lon: 4.872E

Elev: 56

StdP: 100.65

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.7	-2.5	-12.1	1.3	-2.6	-9.4	1.7	-1.2	22.7	3.2	21.0	2.8	5.2	110	0.887

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	20.9	17.0	19.4	16.3	18.1	15.5	17.7	20.1	16.8	18.8	16.0	17.7	5.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	11.9	18.8	15.8	11.3	18.1	15.0	10.7	17.2	49.9	20.1	47.3	18.9	44.9	17.7	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	18.2	16.2	-4.7	23.6	2.3	2.4	-6.3	25.3	-7.7	26.7	-9.0	28.1	-10.7	29.8		
(5)			-6.3	18.5	2.1	1.6	-7.8	19.7	-9.0	20.6	-10.2	21.5	-11.8	22.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	3.3	2.6	3.4	6.0	9.1	11.9	14.5	15.3	13.1	9.5	6.4	4.4	
	DBStd	4.99	2.49	2.62	2.21	2.12	2.45	2.02	2.30	2.12	2.02	2.30	2.61	2.93	
	HDD10.0	1112	206	208	205	123	46	4	0	0	2	35	110	173	
	HDD18.3	3661	464	442	463	371	286	194	121	98	158	273	358	431	
	CDD10.0	502	0	0	0	2	19	59	141	164	94	20	1	0	
	CDD18.3	8	0	0	0	0	0	1	3	4	1	0	0	0	
	CDH23.3	8	0	0	0	0	1	0	5	3	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	8.1	10.2	8.9	8.0	7.1	7.2	6.8	6.5	7.1	8.0	9.0	9.0	9.0	
Precipitation	PrecAvg	1389	140	110	101	79	60	71	90	127	137	157	163	149	
	PrecMax	1646	294	211	183	133	107	152	168	197	214	283	306	281	
	PrecMin	967	27	35	8	33	8	20	41	34	48	53	48	48	
	PrecStd	185	65	45	45	24	24	30	34	35	45	60	61	63	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	7.6	9.6	15.1	19.7	20.8	23.8	22.9	19.6	14.9	12.1	9.9	
		MCWB	7.5	6.7	6.0	10.1	13.3	15.3	17.8	18.0	15.6	13.0	11.1	9.0	
	2%	DB	7.7	6.9	7.2	11.8	16.2	17.8	21.1	20.8	17.8	13.9	11.1	9.2	
		MCWB	6.7	6.2	5.8	8.1	11.8	13.9	17.2	17.5	14.7	12.1	9.9	8.3	
	5%	DB	7.2	6.5	6.5	9.8	14.3	15.9	19.2	19.5	16.6	13.1	10.3	8.7	
		MCWB	6.3	5.8	5.4	7.4	10.7	13.1	16.5	16.7	14.2	11.7	9.1	7.7	
	10%	DB	6.6	6.0	6.0	8.6	12.6	14.6	17.9	18.3	15.8	12.4	9.6	8.0	
		MCWB	5.7	5.4	5.1	6.8	10.0	12.5	15.6	15.9	13.7	11.0	8.4	7.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	7.2	6.8	10.7	14.3	15.9	18.7	19.5	16.6	13.5	11.4	9.2
			MCDB	8.2	7.3	7.6	13.7	18.2	19.1	22.0	21.2	18.4	14.3	11.8	9.5
2%		WB	7.1	6.5	6.2	8.7	12.3	14.5	17.7	18.0	15.5	12.7	10.2	8.7	
		MCDB	7.6	6.8	6.8	10.5	15.3	17.0	20.3	20.3	17.1	13.3	10.9	9.1	
5%		WB	6.6	6.1	5.8	7.8	11.1	13.5	16.8	17.1	14.7	12.1	9.5	8.0	
		MCDB	7.1	6.4	6.4	9.4	13.5	15.4	18.9	19.0	16.2	12.9	10.1	8.6	
10%		WB	6.0	5.5	5.3	7.1	10.2	12.8	15.8	16.2	14.0	11.4	8.7	7.3	
		MCDB	6.5	5.9	5.9	8.2	12.2	14.3	17.6	17.9	15.5	12.3	9.4	8.0	
Mean Daily Temperature Range		5% DB	MDBR	2.5	2.5	2.9	3.4	3.9	3.7	3.5	3.3	3.0	2.6	2.5	2.6
			MCDBR	2.8	2.4	3.2	5.2	6.6	5.3	5.4	4.5	3.6	2.7	2.7	2.7
	MCWBR		3.2	2.6	2.6	3.3	3.7	3.2	3.2	2.8	2.5	2.7	3.2	3.2	
	5% WB	MCDBR	2.7	2.3	2.7	4.5	5.8	4.6	4.9	4.1	3.2	2.6	2.9	2.7	
		MCWBR	3.2	2.6	2.6	3.1	3.6	3.2	3.2	2.9	2.6	3.0	3.5	3.2	
Clear-Sky Solar Irradiance	taub	0.313	0.336	0.354	0.382	0.366	0.362	0.376	0.376	0.361	0.340	0.314	0.290		
	taud	2.162	2.280	2.341	2.310	2.427	2.473	2.451	2.448	2.467	2.398	2.229	2.102		
	Ebn at Noon	461	667	779	819	864	873	849	820	771	662	462	345		
	Edh at Noon	47	70	88	108	103	100	100	93	78	62	44	34		
All-Sky Solar Radiation	RadAvg	0.28	0.84	2.03	3.75	5.17	5.54	4.85	3.74	2.38	1.15	0.39	0.17		
	RadStd	0.04	0.14	0.26	0.34	0.56	0.65	0.46	0.34	0.21	0.14	0.05	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-117	N/A	N/A

Nomenclature: See separate page